Tanmay Patil

MECHANICAL DESIGN | MANUFACTURING | TEST & VALIDATION

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PROFILE

Mechanical engineer combining design, hands-on manufacturing, and troubleshooting to improve how equipment is built, assembled, and maintained.

EXPERIENCE

Design and Rebuild Engineer | Mid-West Forge

May 2025 - Present

- Redesigned a die-box assembly to address recurring bolt shear failures and improve serviceability. Service time fell from 16 hours to 4 hours: a 75% reduction.
- Used RSLinx and RSLogix 500 for PLC troubleshooting. On ABB 6400 robots with S4C controllers, performed rev-counter updates for individual axes and diagnosed pendant wiring, axis-drive, and controller-board problems.
- Currently recreating the gauging screen using EZTouch 2.5 for an EZAutomation HMI.

Engineering Intern | Mid-West Forge

May - August 2024

- Investigated recurring hopper-winch wire-rope failures, performed calculations, and designed a custom damping system.
- Created an interlocking differential display stand in SolidWorks and used DFM / DFA principles for an interlocking aluminum sheet-metal design fabricated with waterjet cutting.

SELECTED ENGINEERING PROJECTS

ATV chain-drive system

The chain-drive system was 25% lighter and took 50% less time to assemble than the previous prop-shaft system.

CVT / ECVT modeling & DAQ

Built a DAQ system to collect sheave linear travel, engine RPM, and wheel-speed RPM, and compared these measurements with a CVT model I developed.

EDUCATION

Master of Science in Mechanical Engineering

May 2025

University of Michigan, Ann Arbor

Bachelor of Science in Mechanical Engineering

May 2023

K.J. Somaiya College of Engineering, Mumbai

TECHNICAL SKILLS

Mechanical design: SolidWorks & PDM, Siemens NX, GD&T, DFM / DFA, Mechanism design, Engineering drawings, Prototyping & fabrication

Manufacturing & controls: Root-cause analysis, Equipment rebuilds, RSLinx / RSLogix 500, EZTouch 2.5, ABB 6400 / S4C, Electrical & pneumatic troubleshooting, PFMEA

Analysis & testing: MATLAB / Simulink, ANSYS FEA, Altair HyperWorks / Inspire, ADAMS / MotionView, Data acquisition, Sensor integration, Model-to-test comparison